# Construction manual

## Version 2 — 12 V supply with brightness control (dimmer)

| | |
|---|---|
| **Part** | Support with light for the pregoscope (telescopic-tube microscope, 40x objective) |
| **Units** | millimetres (mm) · first-angle projection |
| **Tolerance** | ±0.2 mm (ISO 2768-m; PLA, 0.2 mm layer) |
| **Origin** | on the tube axis, z = 0 at the top face of the lid (floor of the box), +z up, +x to the right seen from the front, +y toward the column |
| **Estimated time** | 190 min of assembly, not counting printing |

**Status: pre-calibration**

The dimensions are the design values and have not been checked on printed parts yet. Marked (*) in the drawings.

- Outer tube: the STEP said 30 mm and the caliper 33.5 mm. The 3D and the clamp bore use 33.5 mm; confirm with jig G1.
- Objective: 38 mm total length (manufacturer data) and 35 mm below the inner tube (measured); the real working distance (0.4 to 0.8 mm) depends on the model and must be checked while focusing.
- Cut-outs of the switch, jack and knob, LED flange, slide and screws: confirm with jig G2; the 9 V battery with G3.
- The traces and the solder mask of the boards passed an automatic check but have not been looked at in PCB software.
- The Chinese and English text was written without a technical translator and needs review by a native speaker.

### Revisions

| Rev. | Date | Description |
|---|---|---|
| A | 2026-11-03 | First issue, pre-calibration |

## Before you start

### Tools and materials

- FDM 3D printer, bed of at least 300 mm, PLA filament
- 30 to 40 W soldering iron, 0.8 mm solder, wire stripper
- Multimeter (DC voltage and current)
- Hot-glue gun, heat gun or lighter for the heat-shrink
- Small Phillips screwdriver; 3 mm drill bit; fine sandpaper; caliper

### Safety

- Soldering iron at about 300 °C: use the stand, ventilate the room, never touch the tip.
- Hot glue and heat-shrink burn: let them cool before handling.
- Check the polarity before powering up (LED, diode, battery, positive-centre plug). A reversed connection burns components.
- Use only the 12 V supply listed (v2) and a new 9 V alkaline battery (v1). Do not use rechargeable lithium batteries.
- Do not overtighten screws in PLA and do not force the glass slide.

### Printing

PLA · 0.2 mm · 3 perimeters · 20 to 30 % infill. Dark walls help to reduce stray light.

| ID | Part | Orientation |
|---|---|---|
| M1 | Support: light box, column and arm | file -impressao, lying on its back; supports only inside the box |
| M2 | Tube clamp | file -impressao, opening upward; no supports |
| M3 | Bottom lid | already flat; no supports |
| M4 | Diffuser | flat, natural or white PLA, 1 to 2 layers; no supports |

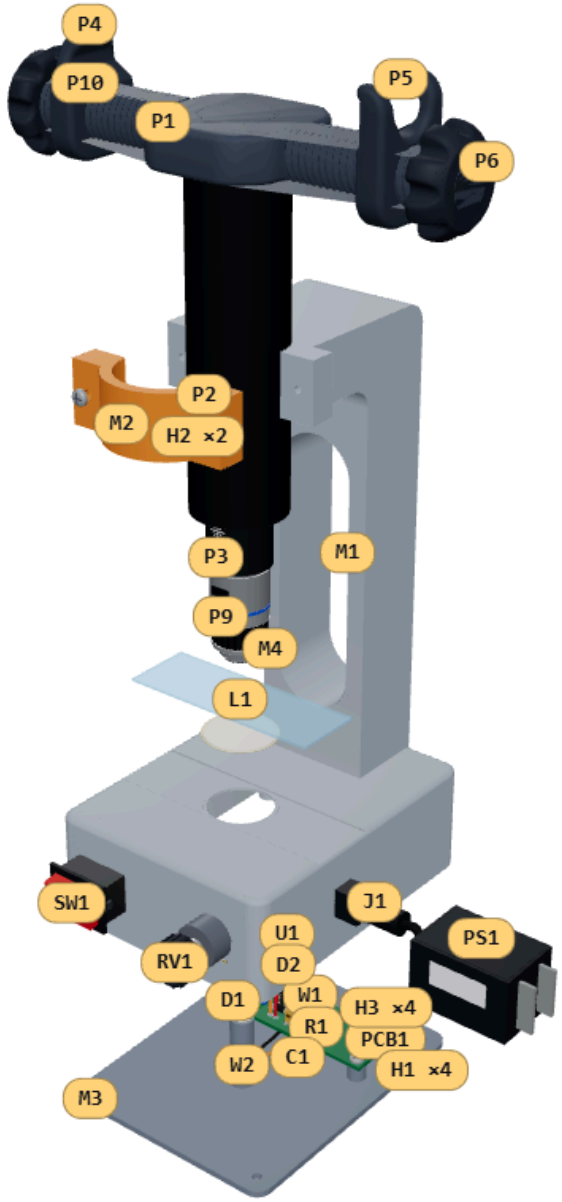
### Files that come with this manual

| File | Content |
|---|---|
| `mecanica/v2-fonte/suporte-base-coluna-impressao.stl` | M1 support, in the print orientation |
| `mecanica/v2-fonte/suporte-base-coluna.stl` | M1 support, in the assembly position |
| `mecanica/comum/abracadeira-tubo-impressao.stl` | M2 clamp, in the print orientation |
| `mecanica/v2-fonte/tampa-fundo.stl` | M3 bottom lid |
| `mecanica/comum/difusor.stl` | M4 diffuser |

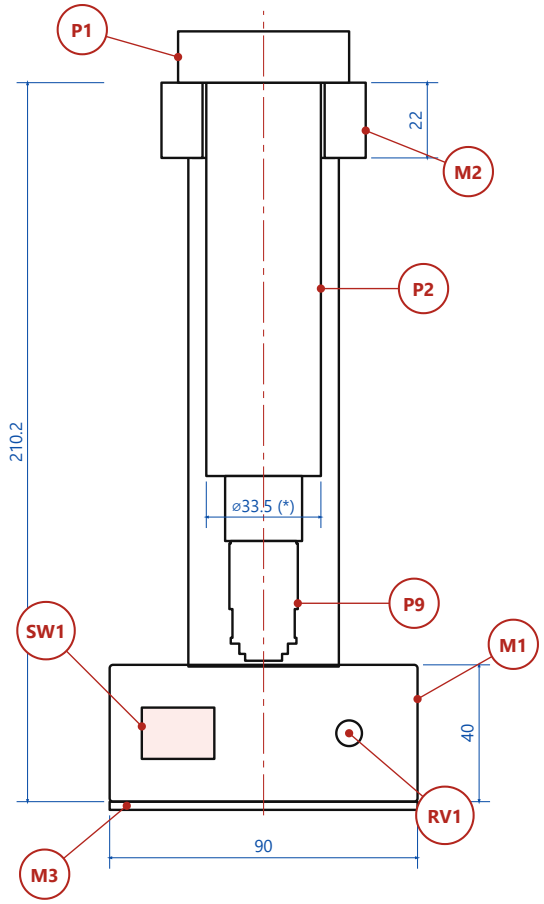
| File | Content |
|---|---|
| `eletronica/v2-fonte-dimmer/esquematico.svg` | Circuit schematic |
| `eletronica/v2-fonte-dimmer/gerber/` | Gerber files of the board PCB1 (fabrication) |
| `eletronica/v2-fonte-dimmer/bom.csv` | Component list of the board |
| `eletronica/v2-fonte-dimmer/pick-and-place.csv` | Component placement coordinates |
| `lista-de-compras-v2.csv` | Shopping list (price to be filled in) |
| `mecanica/calibracao/` | Jigs G1 to G3 (print them first) |

Order of the document: exploded view · 1 technical drawings · 2 positions · 3 bill of materials · 4 wiring · 5 assembly · 6 test · 7 calibration · 8 inspection.

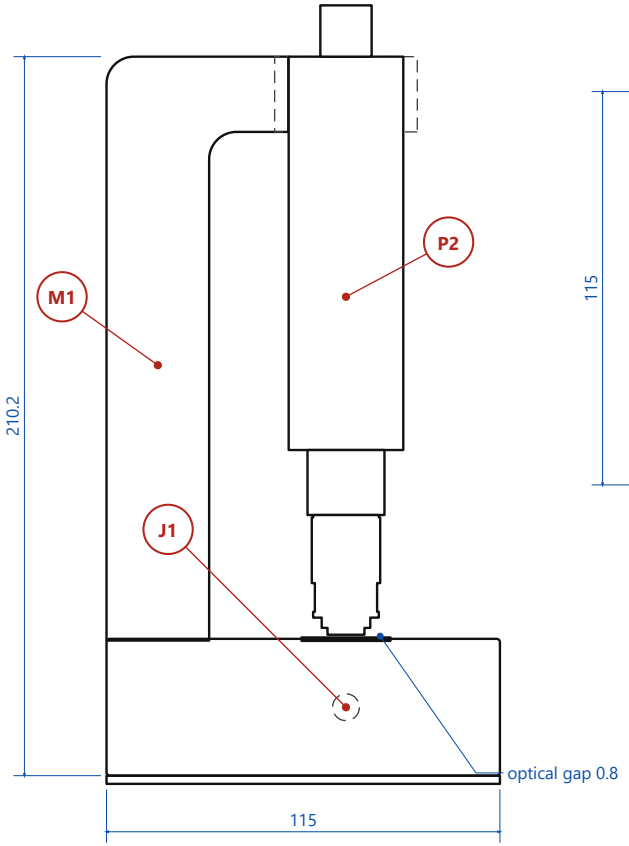
## Exploded view



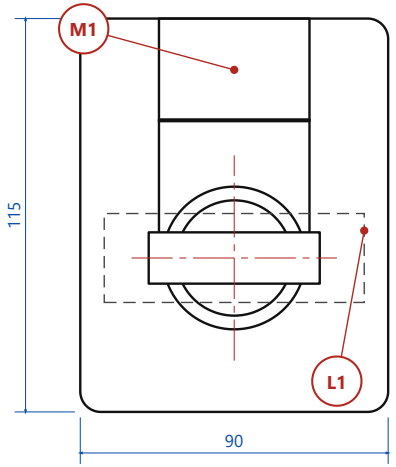

Every part pulled apart with its ID from the bill of materials; “×n” is the quantity.




**Front view**
1:2




**Left view**
1:2



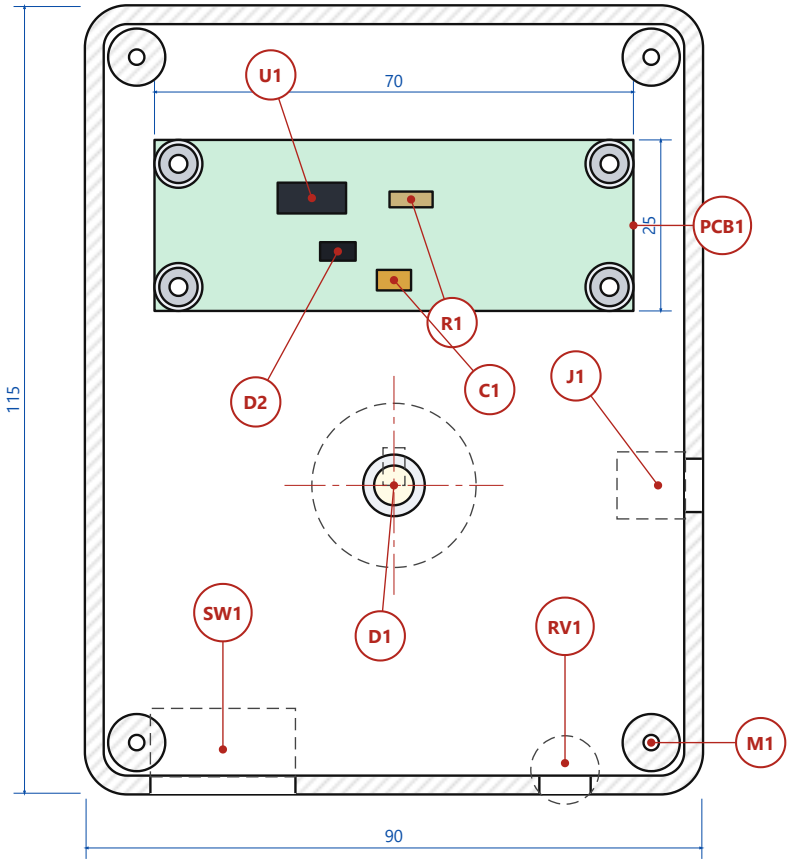

**Top view**
1:2

Tube ø33.5 (*) · sleeve ø22.5 · objective ø20 × 35 · lens at z = 41.2 · gap 0.8

### General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

| | | | |
|---|---|---|---|
| Title: **General assembly** | | | Code: **DT-01-v2** |
| Part: Assembly | Material: PLA | Scale: 1:2 | Sheet: 1 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |




**Plan without lid and ceiling**

1:1

## Centre position of each part (mm)

| ID | X | Y | Z | Size |
|---|---|---|---|---|
| D1 | 0 | 0 | 31.3 | 5.8 × 5.8 × 8.6 |
| PCB1 | 0 | 38 | 6.8 | 70 × 25 × 1.6 |
| U1 | -12 | 42 | 18.4 | 10 × 4.5 × 15.5 |
| R1 | 2.5 | 41.8 | 9 | 6.3 × 2.3 × 2.3 |
| C1 | 0 | 30 | 11.1 | 5 × 2.6 × 5.4 |
| SW1 | -25 | -43.8 | 20 | 21.2 × 10 × 15 |
| H3 | 0 | 38 | 7.6 | 5.6 × 5.6 × 2 |
| W1 | 0 | 8 | 8 | 3 × 20 × 3 |
| W2 | 0 | 8 | 3 | 3 × 10 × 3 |
| J1 | 43.8 | 0 | 20 | 12.4 × 7.8 × 7.8 |
| RV1 | 25 | -43.8 | 20 | 14 × 8 × 14 |
| D2 | -8.2 | 34.2 | 9.2 | 5.2 × 2.7 × 2.7 |

Origin: tube axis, z = 0 at the floor of the box.

PCB1: 70 × 25 × 1.6 mm, top face at z = 7.6.

Wires: see DT-03 and the wiring order (section 4).

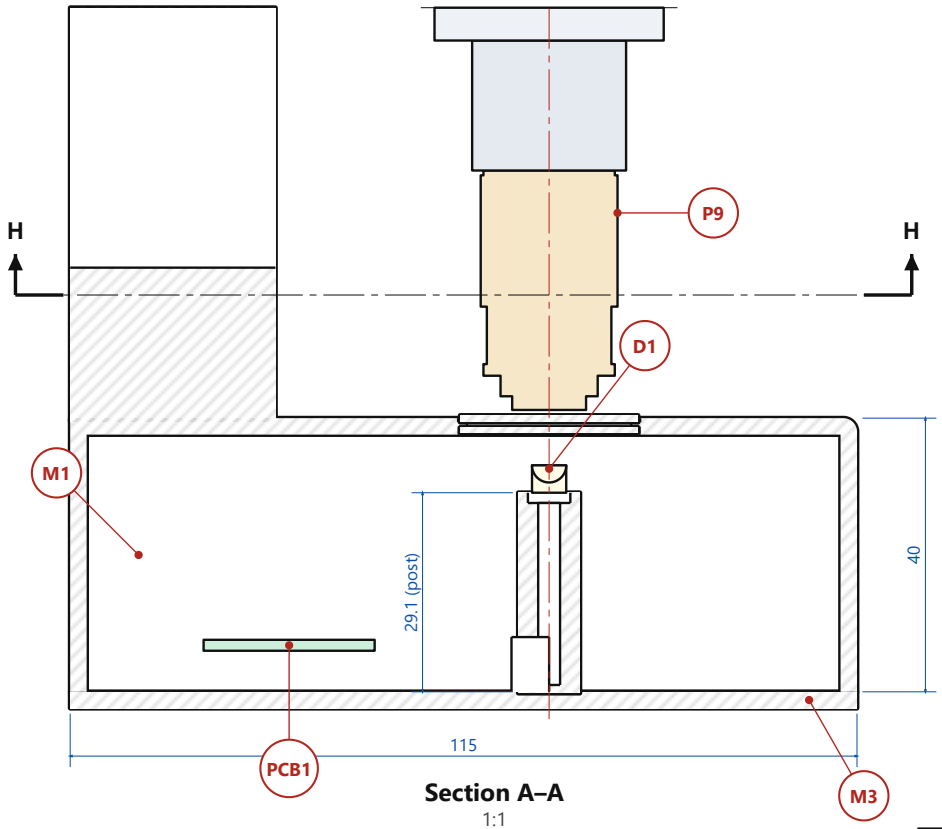
## General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.

2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.

3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

| | | | |
|---|---|---|---|
| Title: **Master plan: inside the box (horizontal section z = 20)** | | | Code: **DT-02-v2** |
| Part: Assembly | Material: PLA | Scale: 1:1 | Sheet: 2 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |

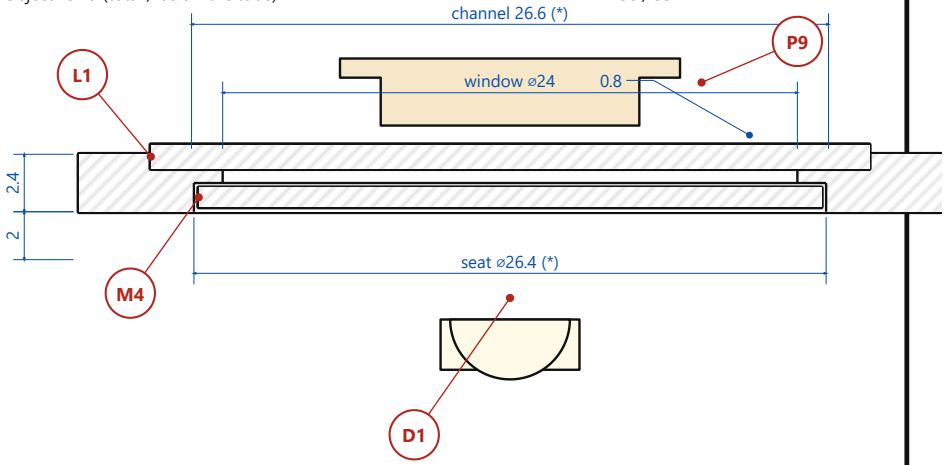



## Section A–A

1:1

### Optical stack (bottom to top)

| | |
|---|---|
| LED D1 (top) | z = 35.6 |
| Diffuser M4 | z = 37.8–38.6 |
| Top face of the ceiling | z = 40 |
| Slide L1 (top) | z = 40.4 |
| Lens of objective P9 | z = 41.2 |
| Working distance | 0.8 mm |
| Objective P9 (total / below the tube) | 38 / 35 mm |



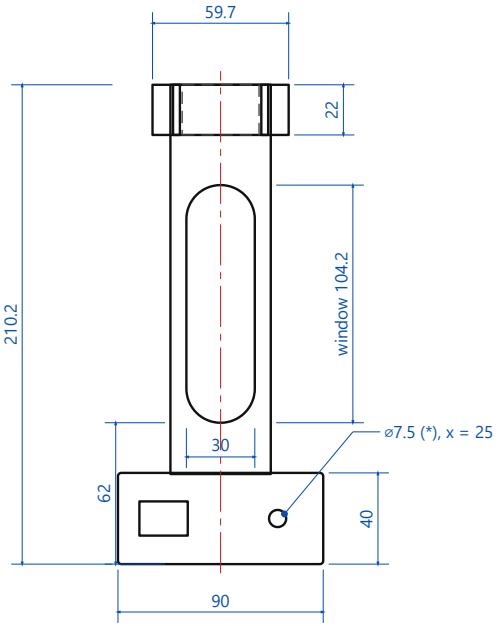

## Detail H: optical stack

3.5:1

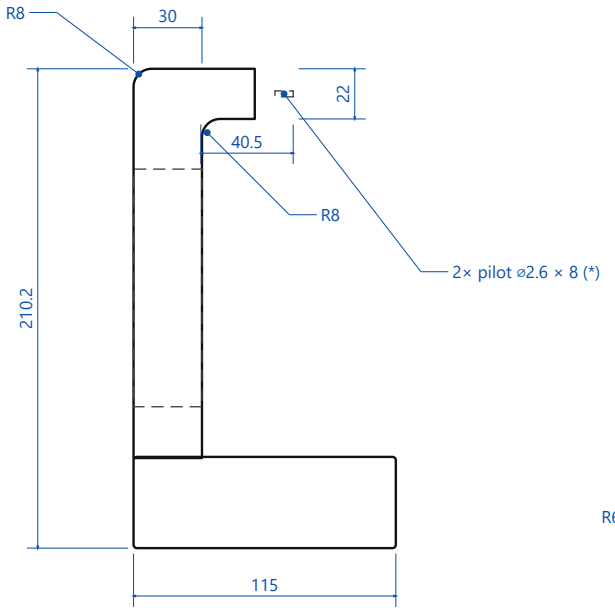
### General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

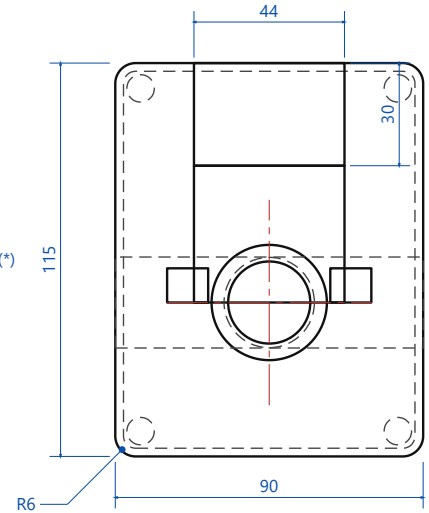
| | | | |
|---|---|---|---|
| Title: **Master plan: section A–A (x = 0), optical stack** | | | Code: **DT-03-v2** |
| Part: Assembly | Material: PLA | Scale: 1:1 · 3.5:1 | Sheet: 3 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |




**Front view**
1:3




**Left view**
1:3




**Top view**
1:2

### Main cut-outs

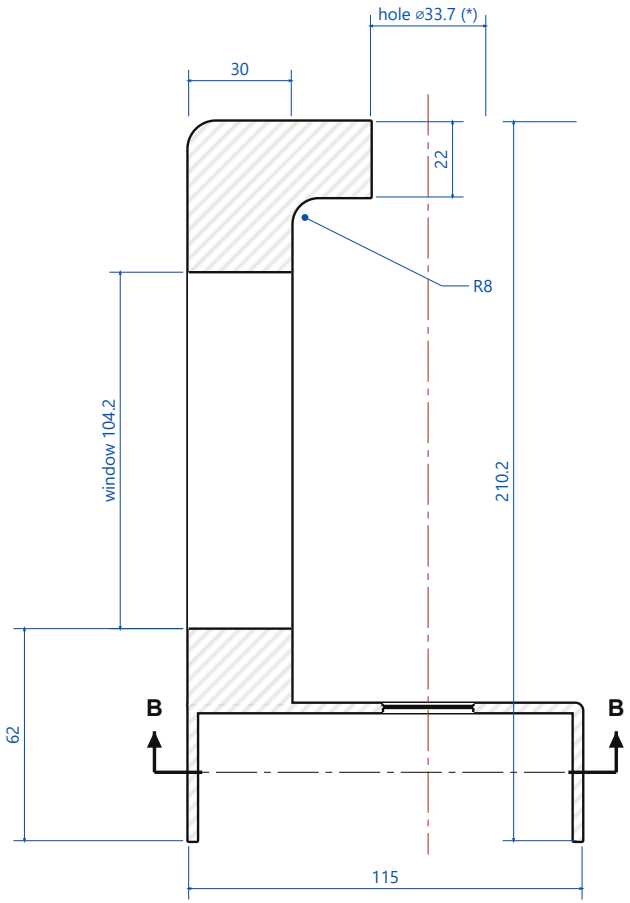
LED window ⌀24 · diffuser seat ⌀26.4 (*) · slide channel 26.6 (*) × 0.6

ring bore ⌀33.7 (*) · column window 30 × 104.2 · cut-out SW1 21.2 × 15 (*)

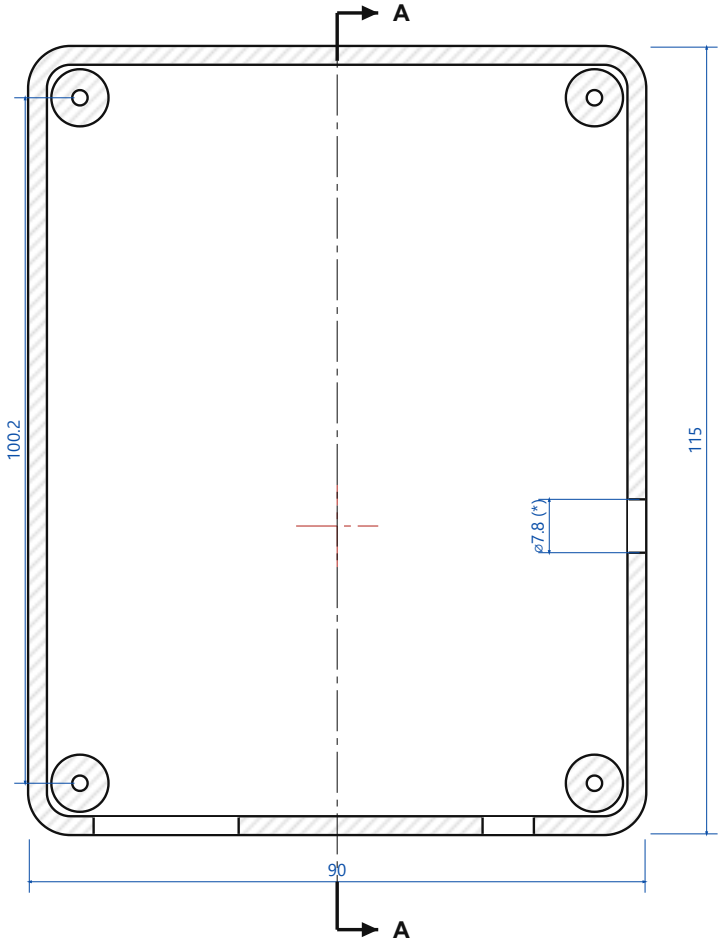
### General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

| | | | |
|---|---|---|---|
| Title: **M1 Support: external views M1** | | | Code: **DT-04-v2** |
| Part: M1 | Material: PLA | Scale: 1:3 · 1:2 | Sheet: 4 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |




**Section A–A (x = 0)**
1:2




**Section B–B (z = 20)**
1:1

Postes: 4× ⌀8, pilot ⌀2.6 (*)
Ceiling, diffuser seat and channel: detail H (DT-03) and detail G (DT-09).

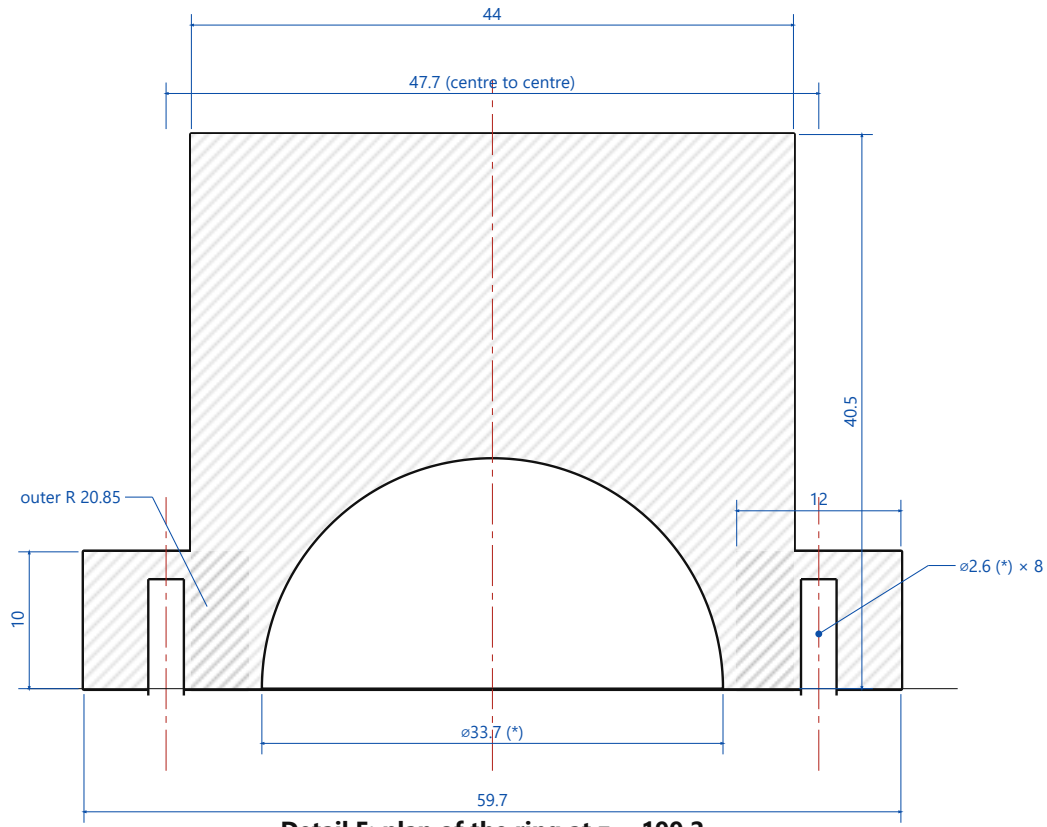
## General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.

2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.

3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

| | | | |
|---|---|---|---|
| Title: **M1 Support: sections A–A and B–B M1** | | | Code: **DT-05-v2** |
| Part: M1 | Material: PLA | Scale: 1:2 · 1:1 | Sheet: 5 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |




**Detail E: plan of the ring at z = 199.2**

2:1

## Table of cut-outs and holes of M1

LED window

⌀24 through the ceiling

(0 ; 0)

Diffuser seat

⌀26.4 (*), from inside the ceiling

(0 ; 0)

Slide channel

26.6 (*) × 0.6 deep, along x

y = 0

SW1 cut-out

21.2 × 15 (*), front wall

x = -25 ; z = 20

J1 hole

⌀7.8 (*), right wall

y = 0 ; z = 20

RV1 hole

⌀7.5 (*), front wall

x = 25 ; z = 20

Column window

30 × 104.2, through along y

z = 62 a 166.2

Lid posts

4× ⌀8, pilot ⌀2.6 (*)

x = ±37.6 ; y = -37.6 e 62.6

Clamp pilot hole

2× ⌀2.6 (*) × 8 deep, along y

x = ±23.85 ; z = 199.2

## Ajustes e folgas

Tube ⌀33.5 in bore ⌀33.7: clearance 0.2

Slide 26 in channel 26.6: clearance 0.6

Diffuser ⌀26 in seat ⌀26.4

M3 screw: through ⌀3.4, pilot ⌀2.6

(*) to be confirmed with jigs G1 to G3.

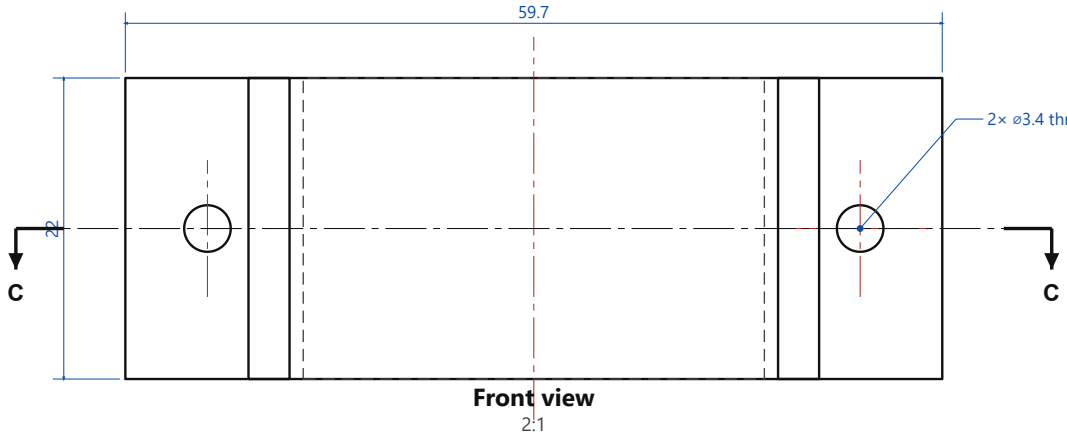
## General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.

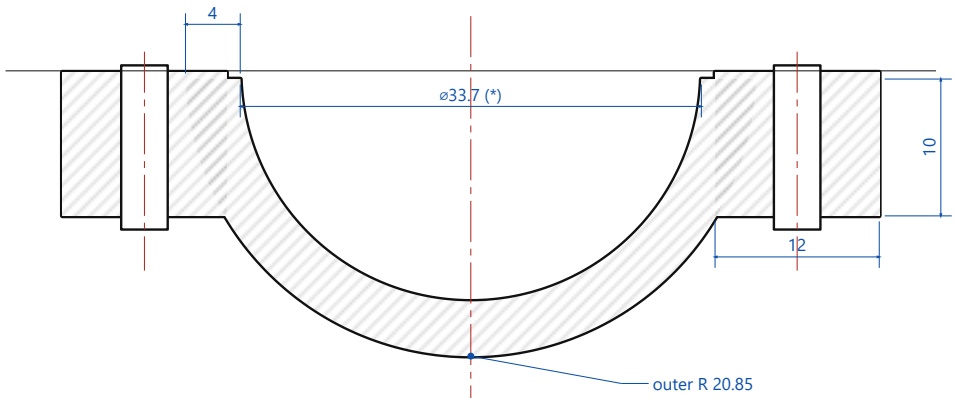
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.

3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

| Title | | | Code |
|---|---|---|---|
| **M1: ring detail and cut-outs M1** | | | **DT-06-v2** |
| Part: M1 | Material: PLA | Scale: 2:1 | Sheet: 6 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |




**Front view**
2:1




**Section C–C (z = 199.2)**
2:1

### Print orientation

File abracadeira-tubo-impressao.stl (opens upward), no supports.

### Assembly

2× screw M3 × 16 (H2) go through the ears and enter 8 mm into the pilot hole ⌀2.6 of M1.

### Tube bore

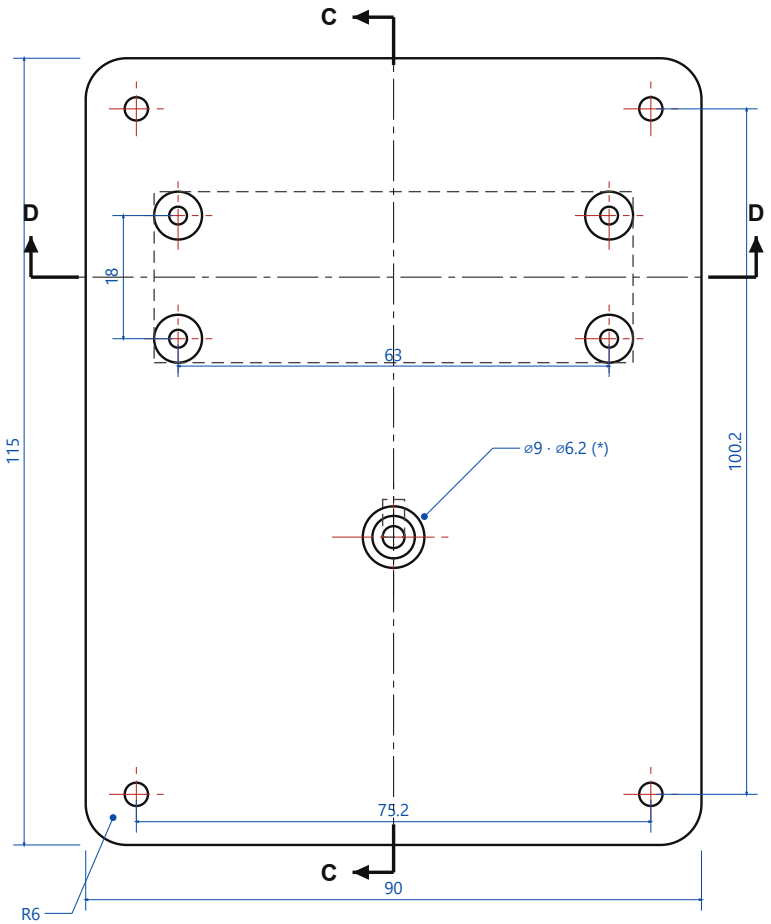
⌀33.7 (*) = measured tube 33.5 + gap 0.2.

Confirm with jig G1 before printing in series.

### General notes

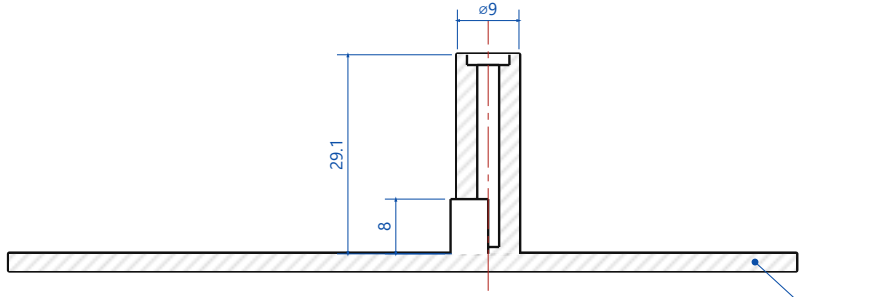
1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

| | | | |
|---|---|---|---|
| Title: **M2 Tube clamp (removable half) M2** | | | Code: **DT-07-v2** |
| Part: M2 | Material: PLA | Scale: 2:1 | Sheet: 7 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |

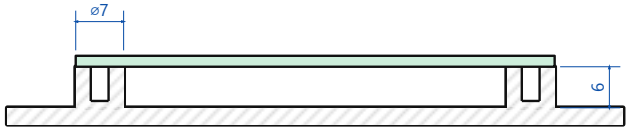



**Top view (print orientation)**
1:1

| | |
|---|---|
| Lid holes (H1) | |
| 4× ø3.4 | x = ±37.6 ; y = -37.6 / 62.6 |
| Board standoffs (H3) | |
| 4× ø7, pilot ø2.6 (*), h = 6 | x = -31.5 / 31.5 ; y = 29 / 47 |
| LED post (D1) | |
| ø9, h = 29.1 | (0 ; 0) |
| Flange seat | |
| ø6.2 (*) × 1.5 | (0 ; 0) |
| Leg hole | |
| ø3.2 blind + slot 3.2 × 5.5 × 8 | slot toward +y |




**Section C–C (x = 0)**
1:1

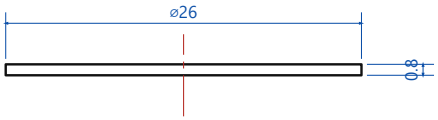



**Section D–D (y = 38)**
1:1

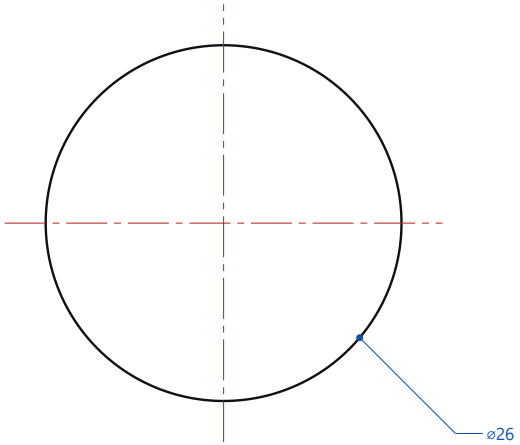
**General notes**

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

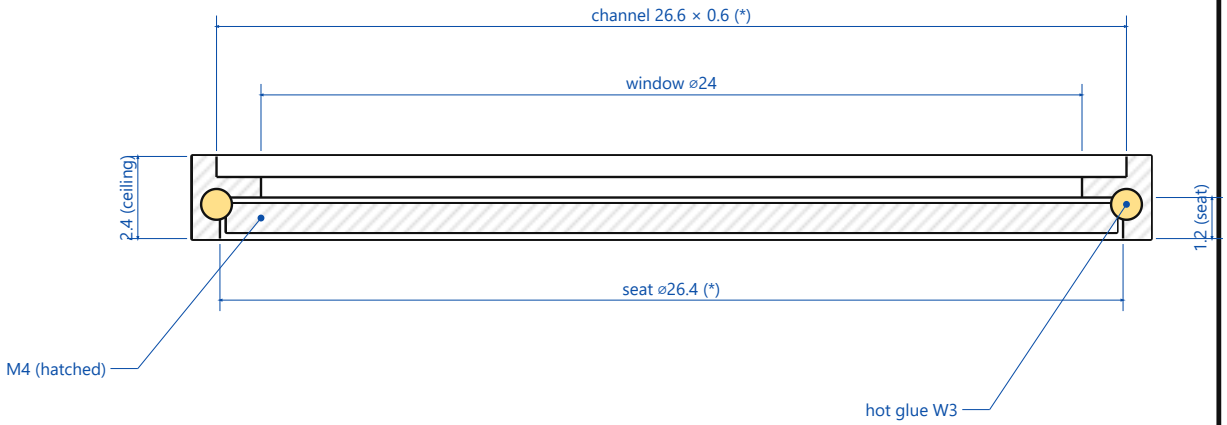
| | | | |
|---|---|---|---|
| Title: **M3 Bottom lid M3** | | | Code: **DT-08-v2** |
| Part: M3 | Material: PLA | Scale: 1:1 | Sheet: 8 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |




**Front view**
2:1




**Top view**
2:1




**Detail G: ceiling of M1 in section**
5:1

### Material and colour

Natural or white PLA, 1 to 2 layers of 0.2 mm (≈ 0.8 mm). No supports.

### General notes

1. Dimensions in mm. General tolerance ±0.2 mm (ISO 2768-m). Fit clearances follow jigs G1 to G3.
2. PLA, 0.2 mm layer, 3 perimeters, 20 to 30 % infill. Print orientation: see the cover.
3. Dimensions marked (*) depend on the physical calibration and may change in revision B.

| | | | |
|---|---|---|---|
| Title: **M4 Diffuser and its seat in the ceiling of M1 M4** | | | Code: **DT-09-v2** |
| Part: M4 | Material: PLA | Scale: 2:1 · 5:1 | Sheet: 9 / 9 |
| Units: mm | Tolerance: ±0.2 (ISO 2768-m) | Revision: A · 2026-11-03 | Version: v2 12 V supply |
| Project: Pregoscope: support with light | | Projection: 1st angle | |

## 2. Position of each part

Centre of the part in the coordinate system of the cover (mm). “Inside” means the part is inside the closed box.

| ID | Name | X | Y | Z | Size X × Y × Z | Fixing | Inside |
|---|---|---|---|---|---|---|---|
| M1 | Support: light box, column and arm | 0 | 12.5 | 105.1 | 90 × 115 × 210.2 | standing on the table | no |
| M3 | Bottom lid | 0 | 12.5 | -1.2 | 90 × 115 × 2.4 | 4 screws H1 at the corners | yes |
| M4 | Diffuser | 0 | 0 | 38.2 | 26 × 26 × 0.8 | hot glue (W3) in the ceiling seat | yes |
| M2 | Tube clamp | 0 | -10.7 | 199.2 | 59.7 × 20.3 × 22 | 2 screws H2 in the ears | no |
| L1 | Microscope slide | 0 | 0 | 39.9 | 76 × 26 × 1 | slid into the channel from the side | no |
| D1 | White 5 mm LED (diffused) | 0 | 0 | 31.3 | 5.8 × 5.8 × 8.6 | seat of the LED post on the lid | yes |
| PCB1 | Driver board | 0 | 38 | 6.8 | 70 × 25 × 1.6 | 4 screws H3 on the 6 mm standoffs | yes |
| U1 | LM317T adjustable regulator | -12 | 42 | 18.4 | 10 × 4.5 × 15.5 | soldered to PCB1, tab facing back | yes |
| R1 | Resistor R1 (sets the current) | 2.5 | 41.8 | 9 | 6.3 × 2.3 × 2.3 | soldered to PCB1 | yes |
| C1 | Ceramic capacitor 100 nF | 0 | 30 | 11.1 | 5 × 2.6 × 5.4 | soldered to PCB1, next to U1 | yes |
| SW1 | Rocker switch KCD1-101 | -25 | -43.8 | 20 | 21.2 × 10 × 15 | front-wall cut-out, snap tabs | no |
| H1 | M3 × 10 self-tapping screw | 0 | 0 | -1.2 | 5.6 × 5.6 × 2 | 4 lid holes (see the table below) | yes |
| H2 | M3 × 16 self-tapping screw | 0 | -12.8 | 199.2 | 54.4 × 16 × 5.6 | 2 clamp ears, Y axis (see the table below) | no |
| H3 | M3 × 8 self-tapping screw | 0 | 38 | 7.6 | 5.6 × 5.6 × 2 | 4 PCB1 holes (see the table below) | yes |
| W1 | Flexible wire 24 AWG | 0 | 8 | 8 | 3 × 20 × 3 | from the LED to PCB1 through the side slot of the LED post | yes |
| W2 | Heat-shrink tube ⌀2-3 mm | 0 | 8 | 3 | 3 × 10 × 3 | over the two LED solder joints | yes |
| W3 | Hot glue and double-sided tape | 0 | 0 | 38 | 26 × 26 × 1 | bead around the diffuser edge | yes |
| P9 | Objective 40x / 0.65 | 0 | 0 | 58.7 | 20 × 20 × 35 | screwed to the inner tube of the pregoscope | no |
| P2 | Outer tube | 0 | 0 | 152.7 | 33.5 × 33.5 × 115 | in the column ring, held by M2 | no |
| P1 | Head: plate and threaded rod | 0 | 0 | 217.7 | 50 × 50 × 15 | resting on top of the ring | no |
| J1 | Panel DC jack DC-022B | 43.8 | 0 | 20 | 12.4 × 7.8 × 7.8 | right-wall hole, nut outside | no |
| RV1 | 220 Ω linear rheostat (WH148 B221) | 25 | -43.8 | 20 | 14 × 8 × 14 | front-wall hole, nut outside | no |

| ID | Name | X | Y | Z | Size X × Y × Z | Fixing | Inside |
|---|---|---|---|---|---|---|---|
| D2 | Diode 1N5819 (reverse polarity) | -8.2 | 34.2 | 9.2 | 5.2 × 2.7 × 2.7 | soldered to PCB1, band toward the pin2 pad | yes |
| PS1 | 12 V 500 mA power adapter | 85 | 0 | 20 | 38 × 26 × 24 | outside the box, cable to the jack J1 | no |

### Screws, one by one

| ID | Screw | X | Y | Z |
|---|---|---|---|---|
| H1 | M3 × 10 self-tapping screw | -37.6 | -37.6 | -2.4 |
| H1 | M3 × 10 self-tapping screw | -37.6 | 62.6 | -2.4 |
| H1 | M3 × 10 self-tapping screw | 37.6 | -37.6 | -2.4 |
| H1 | M3 × 10 self-tapping screw | 37.6 | 62.6 | -2.4 |
| H3 | M3 × 8 self-tapping screw | -31.5 | 29 | 6 |
| H3 | M3 × 8 self-tapping screw | -31.5 | 47 | 6 |
| H3 | M3 × 8 self-tapping screw | 31.5 | 29 | 6 |
| H3 | M3 × 8 self-tapping screw | 31.5 | 47 | 6 |
| H2 | M3 × 16 self-tapping screw | -23.8 | -12.8 | 199.2 |
| H2 | M3 × 16 self-tapping screw | 23.9 | -12.8 | 199.2 |

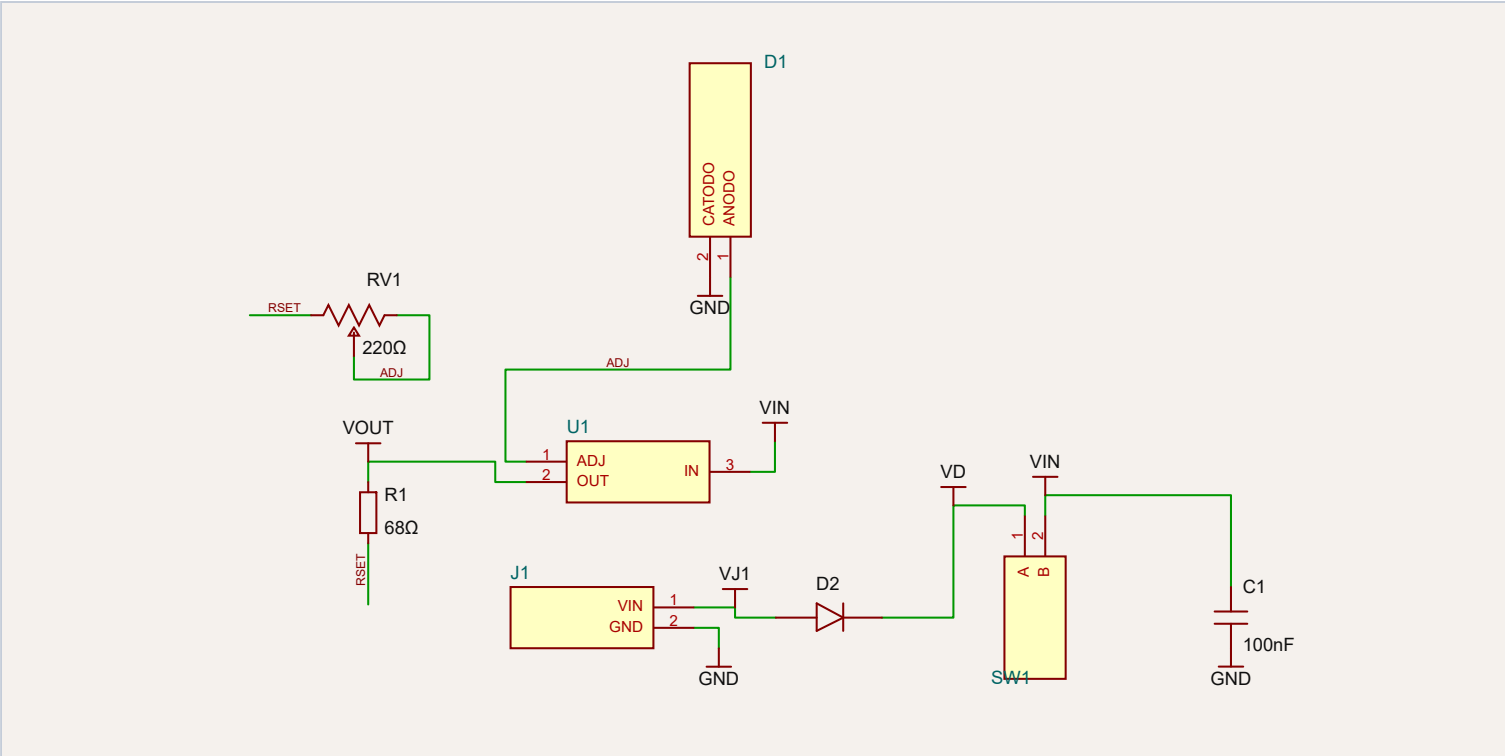
## 3. Bill of materials (BOM)

| ID | Name | Qty | Standard size | Type |
|---|---|---|---|---|
| M1 | Support: light box, column and arm | 1 | PLA, base 90 × 115 × 40 mm, 210,2 mm de altura, cantos de 6 mm, janela de alívio de 30 mm na coluna | printed |
| M2 | Tube clamp | 1 | Anel de 22 mm de altura, furo de 33,7 mm, dois parafusos M3 × 16 | printed |
| M3 | Bottom lid | 1 | Placa de 2,4 mm com porta-LED; v1 leva o berço da pilha, v2 leva 4 suportes da placa do dimmer | printed |
| M4 | Diffuser | 1 | Disco de ⌀ 26 × 0,8 mm em PLA natural ou branco | printed |
| P1 | Head: plate and threaded rod | 1 | 50 × 15 mm, haste de 170 mm | supplied (pregoscope) |
| P2 | Outer tube | 1 | tube ⌀ 33.5 mm × 115 mm (measured) | supplied (pregoscope) |
| P3 | Inner tube (telescopic) | 1 | tube ⌀ 22.5 mm, pulls out up to 40 mm, printed mm scale | supplied (pregoscope) |
| P4 | Left jaw | 1 | 10 mm de espessura | supplied (pregoscope) |
| P5 | Right jaw | 1 | 10 mm de espessura | supplied (pregoscope) |
| P6 | Right knob | 1 | ⌀ 34 mm | supplied (pregoscope) |
| P10 | Left knob (mirror of P6) | 1 | ⌀ 34 mm, igual à P6 | supplied (pregoscope) |
| P9 | Objective 40x / 0.65 | 1 | objective 40x / 0.65 (160/0.17, RMS thread), ⌀ 20 mm, 38 mm long (35 mm below the inner tube) | purchased |
| L1 | Microscope slide | 1 | 76 × 26 × 1 mm (microscopy standard) | purchased |
| D1 | White 5 mm LED (diffused) | 1 | ⌀ 5.0 mm; flange ⌀ 5.8 mm; height ≈ 8.6 mm; lead pitch 2.54 mm | purchased |
| R1 | Resistor R1 (sets the current) | 1 | ¼ W axial: body ≈ 6.3 × 2.3 mm; lead ⌀ 0.5 mm; pitch 10 mm | purchased |
| C1 | Ceramic capacitor 100 nF | 1 | ceramic 100 nF (104) 50 V; pitch 5 mm | purchased |
| D2 | Diode 1N5819 (reverse polarity) | 1 | DO-41: body ≈ 5.2 × 2.7 mm; lead ⌀ 0.8 mm; band = cathode | purchased |
| SW1 | Rocker switch KCD1-101 | 1 | panel cut-out 21.2 × 15.0 mm; 2 terminals | purchased |
| PS1 | 12 V 500 mA power adapter | 1 | 12 V 500 mA; P4 plug 5.5 × 2.1 mm, centre positive | purchased |
| J1 | Panel DC jack DC-022B | 1 | DC-022B: panel hole ⌀ 7.8 mm; plug 5.5 × 2.1 mm | purchased |
| U1 | LM317T adjustable regulator | 1 | TO-220: 10.0 × 4.5 × 15.0 mm; pins ADJ-OUT-IN, pitch 2.54 mm | purchased |
| RV1 | 220 Ω linear rheostat (WH148 B221) | 1 | WH148: shaft ⌀ 6 mm; M7 thread; hole ⌀ 7.5 mm | purchased |
| PCB1 | Driver board | 1 | 1.6 mm thick; M3 holes 3.5 mm from the edges | purchased |

| ID | Name | Qty | Standard size | Type |
|---|---|---|---|---|
| H1 | M3 × 10 self-tapping screw | 4 | M3 × 10; pan head ⌀ ≈ 5.6 mm | purchased |
| H2 | M3 × 16 self-tapping screw | 2 | M3 × 16; pan head ⌀ ≈ 5.6 mm | purchased |
| H3 | M3 × 8 self-tapping screw | 4 | M3 × 8; pan head ⌀ ≈ 5.6 mm | purchased |
| W1 | Flexible wire 24 AWG | 1 | 24 AWG: conductor ⌀ 0.51 mm; insulation ⌀ ≈ 1.4 mm | purchased |
| W2 | Heat-shrink tube ⌀2-3 mm | 1 | ⌀ 2 to 3 mm before shrinking; 10 mm pieces | purchased |
| W3 | Hot glue and double-sided tape | 1 | glue stick ⌀ 7 mm; 12 mm double-sided tape | purchased |

The sizes are the catalogue ones; check the real parts with jigs G1 to G3 before printing in series. Parts P1 to P10 are the pregoscope itself and are supplied by the customer.

## 4. Schematic and wiring




### Soldering order on the board

1. Solder U1 (label facing forward: ADJ, OUT, IN), R1 (68 Ω), C1 (100 nF) and D2 (1N5819, band toward the pin2 pad) to the pads of the board. The traces make the connections between them.
2. LED D1: orange wire (anode) to pad D1 ANODO; black wire (cathode) to pad D1 CATODO.
3. Jack J1: red wire (plug centre, positive) to pad J1 VIN; black wire to pad J1 GND. Confirm the polarity with the multimeter.
4. Switch SW1: terminal 1 (red wire) to pad SW1 A; terminal 2 (orange wire) to pad SW1 B.
5. Rheostat RV1: yellow wire to pad RV1 pin1, green wire (wiper) to pad pin2 and blue wire to pad pin3.
6. Cover the LED joints with heat-shrink and tie down loose wires with a dab of hot glue.

### Circuit connections (reference for perforated board)

1. Supply PS1 (12 V) into jack J1. Confirm with the multimeter that the plug centre is positive.
2. Positive of J1 to the anode of D2 (1N5819, the side without a band). Cathode of D2 (band side) to terminal 1 of SW1.
3. Terminal 2 of SW1 to pin IN of U1. C1 (100 nF) between IN and GND, close to U1.
4. Pin OUT of U1 to R1 (68 Ω); R1 to terminal 1 of RV1; terminals 2 and 3 of RV1 tied together.
5. Terminals 2 and 3 of RV1 to pin ADJ of U1 and to the anode of D1.

6. Cathode of D1 to GND of J1. The other terminal of C1 also goes to GND.

### Wire cut list

| ID | Colour | Free end | Board pad | Path (mm) | Cut to (mm) |
|---|---|---|---|---|---|
| W1-1 | orange | D1 A | D1 ANODE | 56 | **85** |
| W1-2 | black | D1 C | D1 CATHODE | 56 | **85** |
| W1-3 | red | J1 + | J1 VIN | 62 | **145** |
| W1-4 | black | J1 - | J1 GND | 65 | **150** |
| W1-5 | red | SW1 1.l | SW1 A | 67 | **150** |
| W1-6 | orange | SW1 2.l | SW1 B | 64 | **150** |
| W1-7 | yellow | RV1 GND | RV1 pin1 | 84 | **170** |
| W1-8 | green | RV1 SIG | RV1 pin2 | 85 | **170** |
| W1-9 | blue | RV1 VCC | RV1 pin3 | 76 | **160** |

Cut = path + 12 mm of stripped ends (6 mm each side) + slack (15 mm; 70 mm on the lid wires, so the lid opens without disconnecting anything). 24 AWG flexible wire.

Board orientation: the "FRENTE / FRONT" arrow on the silkscreen points to the front of the box (switch side); mount the board with the arrow that way.

### Circuit board (Gerber files)

Top side (components)

Bottom side (traces)

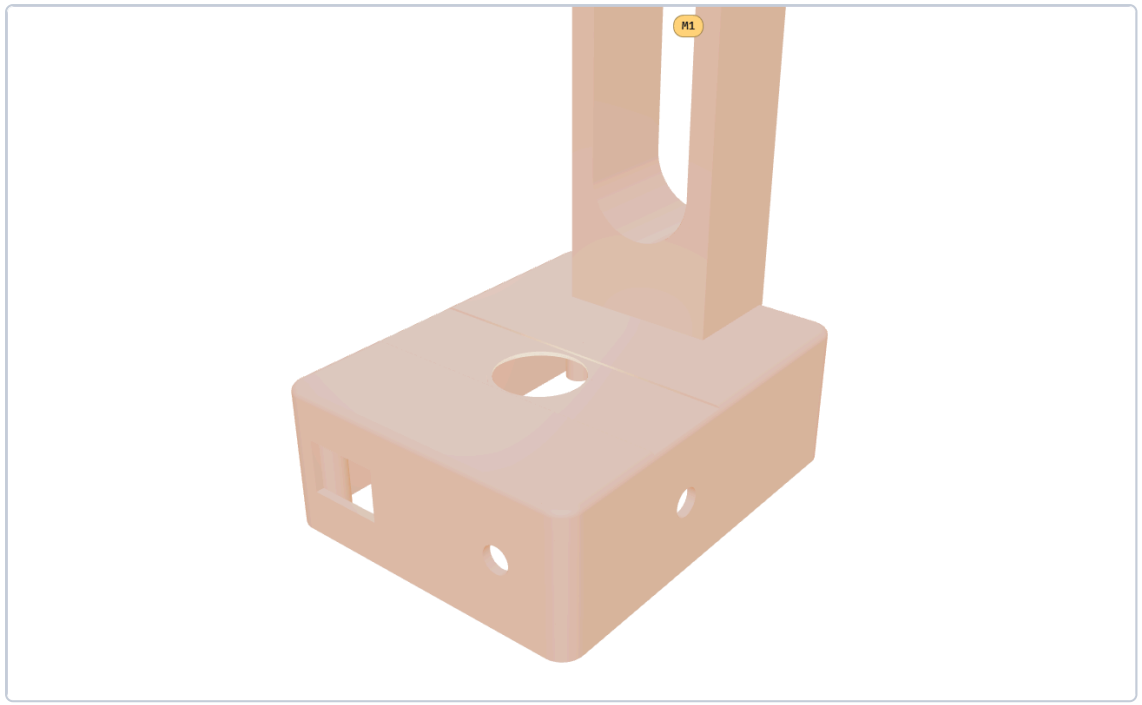
Drawn from the Gerber files themselves with a program independent of the one that made them. The copper connectivity was checked against the schematic: no open traces and no shorts.

Red wire = positive, black = GND. The LM317 holds 1.25 V across R1, so the LED current stays fixed.

1

## Prepare the box (M1)

20 min




Remove the print supports and clean every hole with a 3 mm drill bit. Sand the edges of the slide channel lightly and make sure no debris is left inside the box.

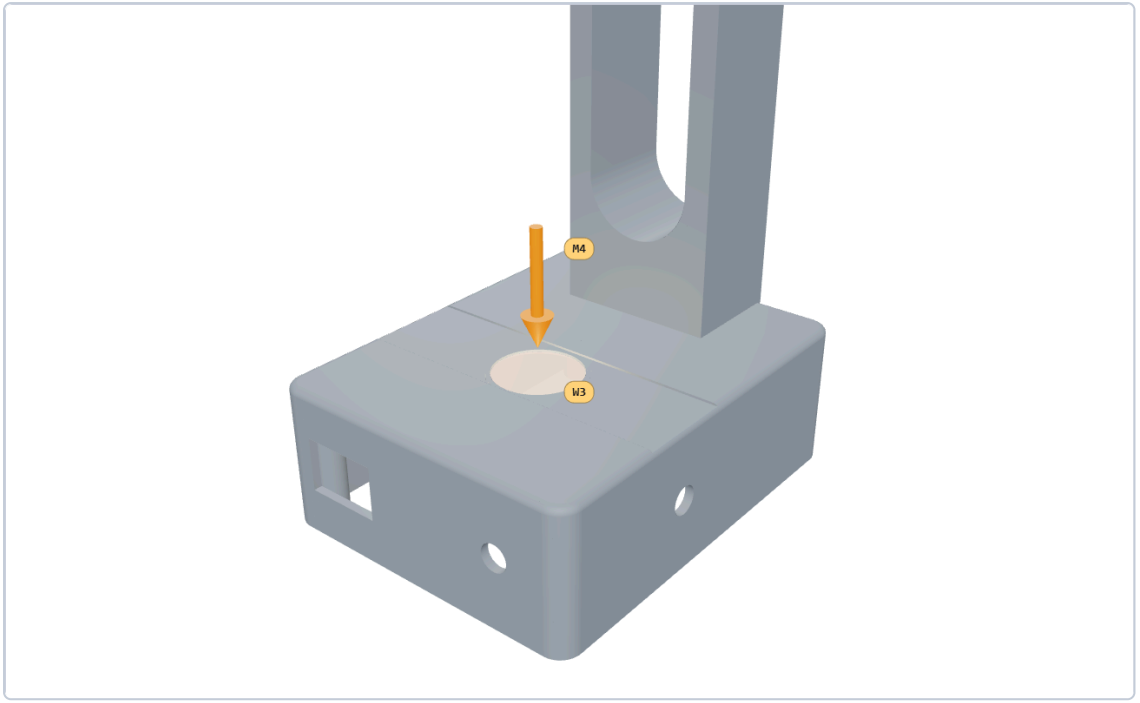
✓
The slide channel must be smooth and clean.

| ID | Part in this step | Qty |
|---|---|---|
| M1 | Support: light box, column and arm | 1 |

2

## Glue the diffuser (M4)

10 min




Place the disc M4 (⌀ 26 × 0.8 mm) in the ceiling seat from inside the box. Fix it with a thin bead of hot glue (W3) around the edge, without covering the ⌀ 24 mm window.

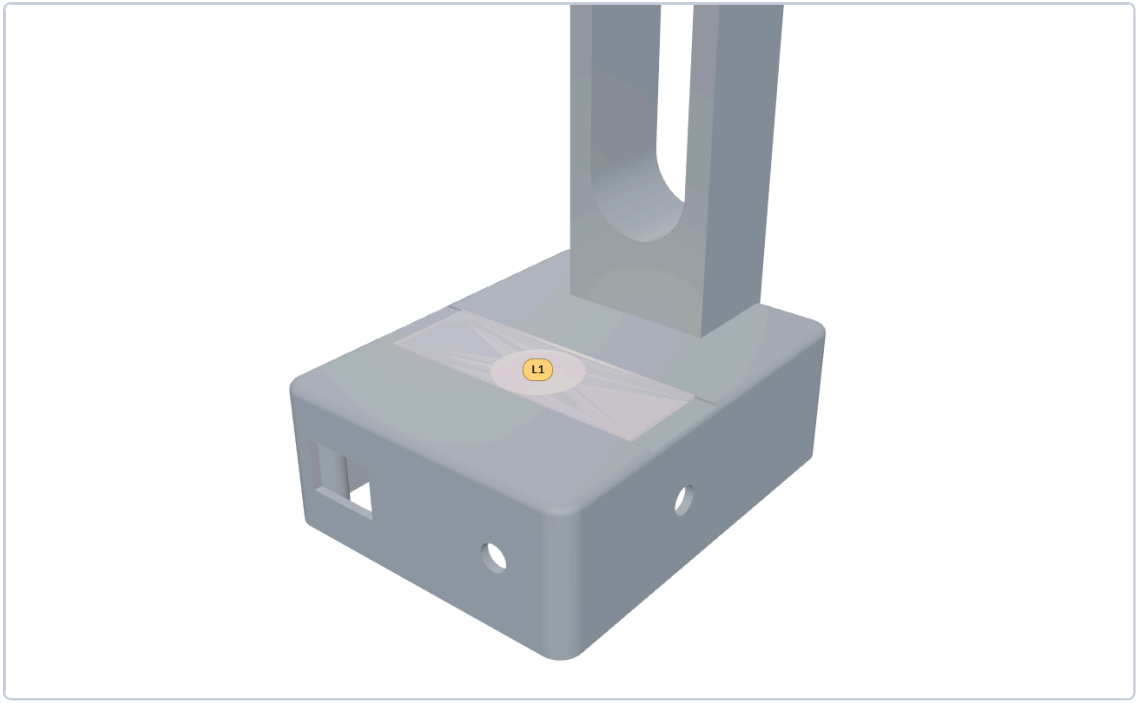
⚠
Hot glue burns: use the tip of the gun and let it cool for 1 minute.

| ID | Part in this step | Qty |
|---|---|---|
| M4 | Diffuser | 1 |
| W3 | Hot glue and double-sided tape | 1 |

3

## Test-fit the slide (L1)

5 min




Slide the slide L1 (76 × 26 × 1 mm) through the channel. It must go in and out without effort and without visible side play. If it sticks, use jig G2 to pick the right size before printing again.

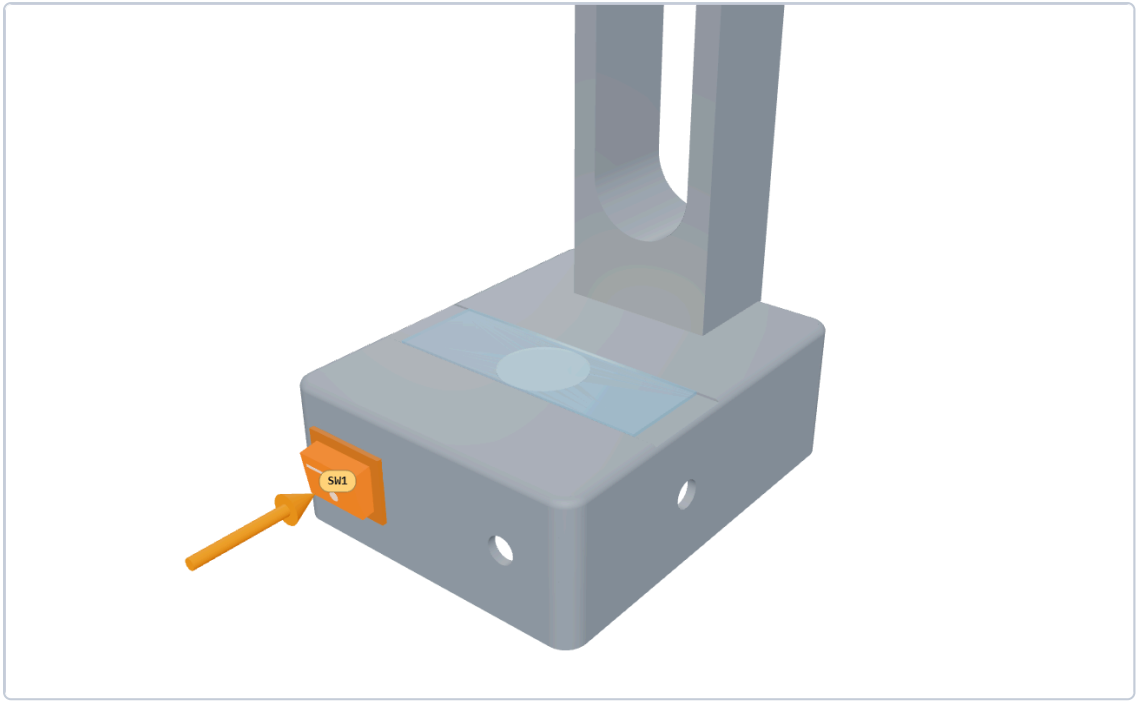
✓
The slide enters from the side and sits 40.4 mm above the floor.

| ID | Part in this step | Qty |
|---|---|---|
| L1 | Microscope slide | 1 |

4

## Install the switch (SW1)

5 min



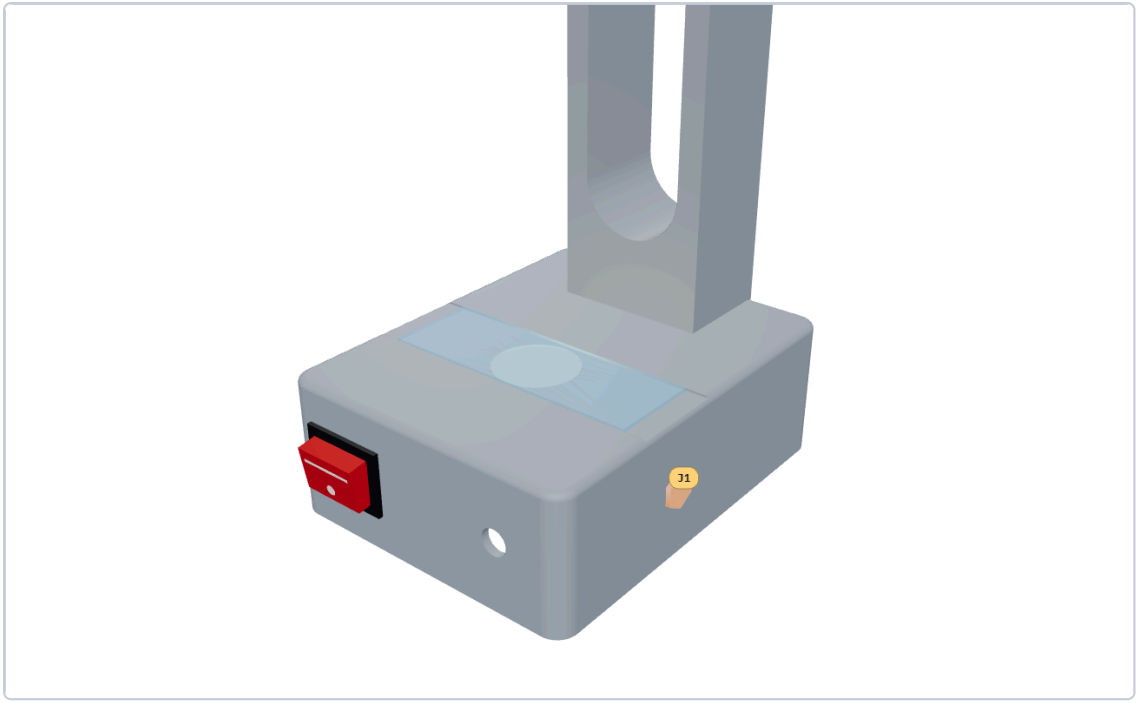

Push the KCD1-101 switch from outside into the 21.2 × 15.0 mm cut-out of the front wall until the tabs click. The two terminals end up inside the box.

| ID | Part in this step | Qty |
|---|---|---|
| SW1 | Rocker switch KCD1-101 | 1 |

5

## Install the power jack (J1)

5 min




| ID | Part in this step | Qty |
|---|---|---|
| J1 | Panel DC jack DC-022B | 1 |

Pass the DC-022B jack from inside through the ⌀ 7.8 mm hole of the right wall and tighten the nut by hand outside. If the hole is loose or tight, check jig G2.

6

## Install the rheostat (RV1)

5 min



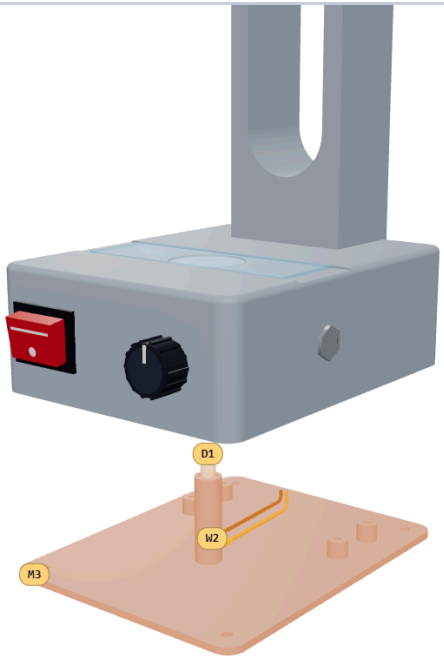

Pass the shaft of the 220 Ω rheostat through the ⌀ 7.5 mm hole of the front wall from inside. Tighten the nut outside and fit the knob.

| ID | Part in this step | Qty |
|---|---|---|
| RV1 | 220 Ω linear rheostat (WH148 B221) | 1 |

7

## Lid: mount the LED (M3, D1)

20 min




Solder two wires (W1) to the LED legs: orange to the long leg (anode), black to the short one (cathode). Insulate each joint with heat-shrink (W2). Seat the LED in the LED post of lid M3. The wires go down the hole of the post and leave through the side slot, which faces the board.

⚠
Solder each leg for less than 3 seconds so the LED does not burn out. Soldering iron at 300 °C: rest it in its stand and never touch the tip. Solder in a ventilated place.

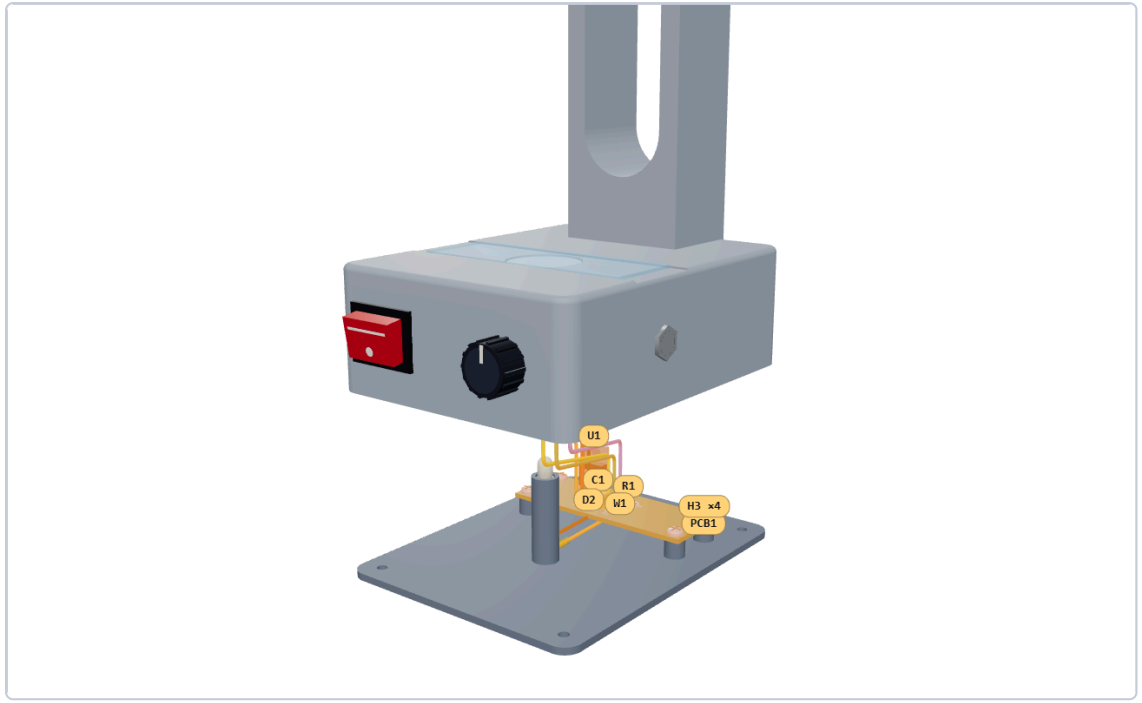
✓
Long leg = anode (+). The lid has no hole through to the outside.

| ID | Part in this step | Qty |
|---|---|---|
| M3 | Bottom lid | 1 |
| D1 | White 5 mm LED (diffused) | 1 |
| W1 | Flexible wire 24 AWG | 1 |
| W2 | Heat-shrink tube ⌀2-3 mm | 1 |

8

## Lid: driver board (PCB1, U1, R1, C1, D2)

40 min




Solder the regulator U1, the resistor R1 (68 Ω), the capacitor C1 (100 nF) and the diode D2 (band toward the pin2 pad) to PCB1 (70 × 25 mm), on the pads of the Gerber files. U1 faces forward with its label: ADJ, OUT, IN from left to right. Screw the board to the 4 standoffs of the lid with M3 × 8 (H3). Solder the wires of the LED and of the other parts to the marked pads, in the order of the wiring section.

⚠
Soldering iron at 300 °C: rest it in its stand and never touch the tip. Solder in a ventilated place.

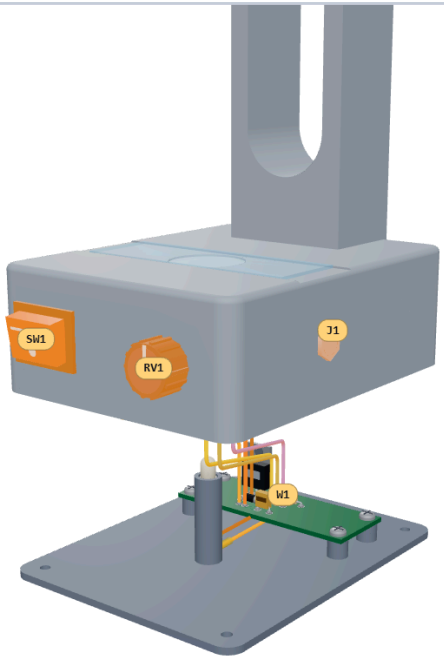
✓
Tab of U1 facing back; no solder joint touches another.

| ID | Part in this step | Qty |
|---|---|---|
| PCB1 | Driver board | 1 |
| U1 | LM317T adjustable regulator | 1 |
| R1 | Resistor R1 (sets the current) | 1 |
| C1 | Ceramic capacitor 100 nF | 1 |
| D2 | Diode 1N5819 (reverse polarity) | 1 |
| H3 | M3 × 8 self-tapping screw | 4 |

9

## Connect the panel wires

30 min



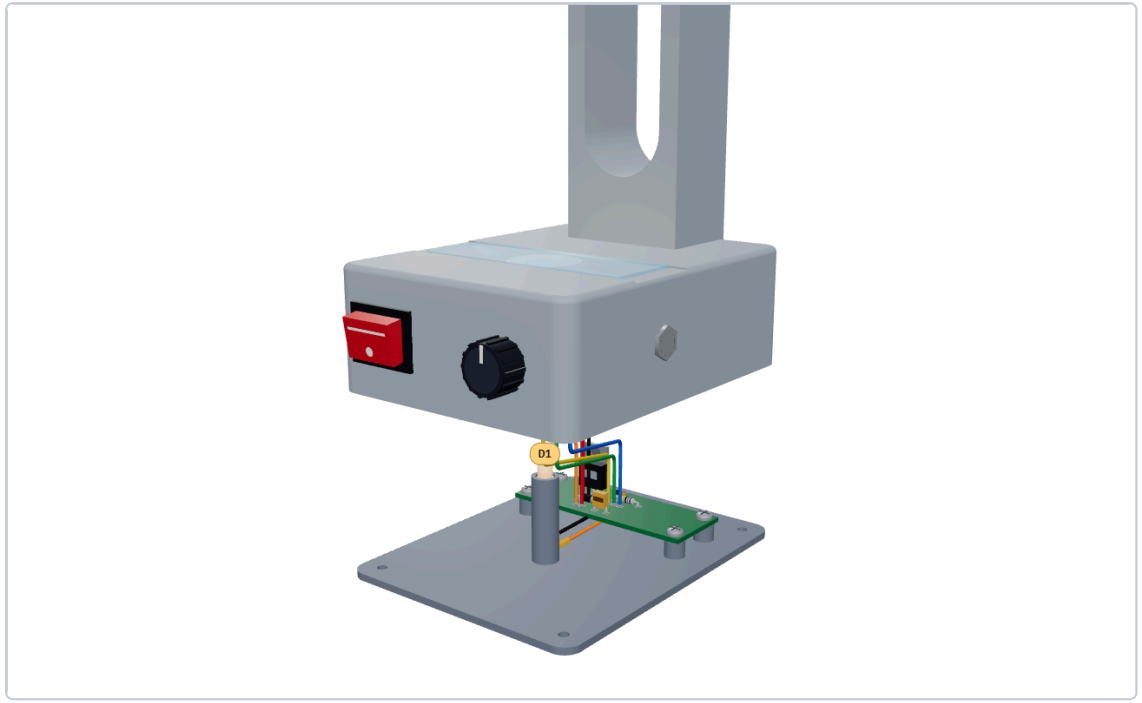

Solder the wires of the jack J1 (pads VIN and GND), of the switch SW1 (pads A and B) and of the three terminals of the rheostat RV1 (pads pin1, pin2 and pin3) in the colours of the wiring section (red positive, black GND, yellow, green and blue for the rheostat). Leave slack so the lid can open.

⚠
Soldering iron at 300 °C: rest it in its stand and never touch the tip. Solder in a ventilated place.

10

## Electrical test with the multimeter

15 min




Before closing, switch the power on and follow the test plan of the electrical test section point by point. Every reading must be inside its range and the LED must light. If a reading is out of range, stop and fix it before going on.

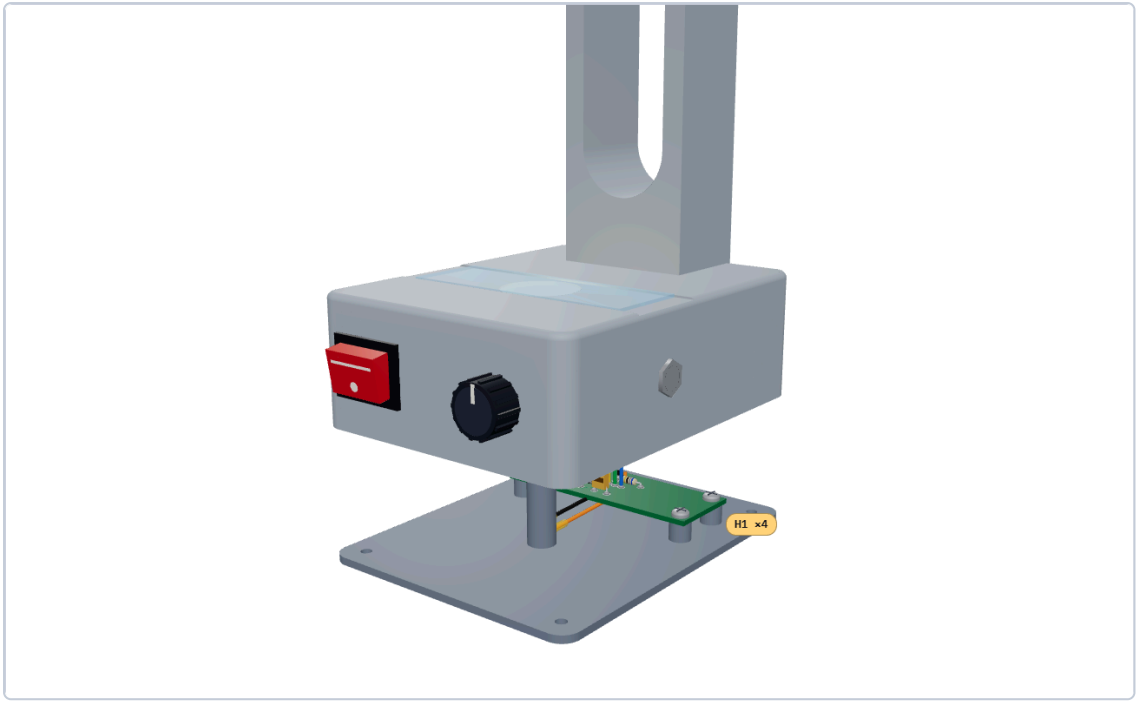
⚠
Check the polarity before powering up: LED (long leg +), diode (band = cathode), plug with positive centre.

✓
18.4 to 4.3 mA in the LED

11

## Close the lid (H1)

10 min




Raise the lid to the box without trapping any wire and tighten the 4 screws M3 × 10 (H1) at the corners. Tighten a little at a time, in a cross pattern, without forcing the plastic.

⚠
Do not overtighten: PLA cracks. Stop when the screw seats.

| ID | Part in this step | Qty |
|---|---|---|
| H1 | M3 × 10 self-tapping screw | 4 |

12

## Fit the pregoscope (M2, H2, P1 to P10)

15 min




Take off the removable half of the clamp M2, lean the outer tube P2 into the column ring, put the half back and tighten the 2 screws M3 × 16 (H2) without forcing. The head P1 rests on top of the ring and the objective P9 sits over the window.

⚠
Do not overtighten: PLA cracks. Stop when the screw seats.

| ID | Part in this step | Qty |
|---|---|---|
| M2 | Tube clamp | 1 |
| H2 | M3 × 16 self-tapping screw | 2 |
| P1 | Head: plate and threaded rod | 1 |
| P2 | Outer tube | 1 |
| P3 | Inner tube (telescopic) | 1 |
| P4 | Left jaw | 1 |
| P5 | Right jaw | 1 |

| ID | Part in this step | Qty |
|---|---|---|
| P6 | Right knob | 1 |
| P10 | Left knob (mirror of P6) | 1 |
| P9 | Objective 40x / 0.65 | 1 |

13

## Connect the supply, focus and brightness (PS1)

10 min




Plug the 12 V supply (PS1) into the jack J1 and switch SW1 on. Turn the knob RV1 to set the brightness. Loosen the clamp and slide the outer tube, or pull the inner tube P3, until the slide is sharp. Nominal focus: the lens 0.8 mm from the slide.

| ID | Part in this step | Qty |
|---|---|---|
| PS1 | 12 V 500 mA power adapter | 1 |

## 6. Electrical test

Supply: the 12 V adapter, white LED of 3.1 V. Measure in this order.

| Measuring point | Expected | Range | If out of range | Measured |
|---|---|---|---|---|
| Supply at J1: positive to negative, no load | 12.00 V | 11.5 to 15 V | An unregulated supply can read up to 15 V with no load; below 11.5 V the supply is weak. | |
| After D2 to GND | 11.60 V | 11 to 11.8 V | Diode reversed (reads 0 V) or drop above 0.5 V (wrong diode). | |
| Input of U1 (after SW1) to GND | 11.60 V | 10.9 to 11.8 V | Switch off or cold joint. | |
| Output of U1 (OUT) to GND | 4.35 V | 4.1 to 4.6 V | R1 or RV1 open: the output rises close to the input. | |
| Anode of D1 (pin ADJ) to GND | 3.10 V | 2.8 to 3.4 V | LED reversed or burnt out. | |
| Across R1 + RV1 (between OUT and ADJ) | 1.25 V | 1.2 to 1.3 V | Outside 1.25 V the LM317 is fake or out of regulation. | |
| LED current with RV1 at minimum (0 Ω) | 18.4 mA | 17 to 19.4 mA | Wrong R1 (68 Ω) or the wiper of RV1 is not tied to pins 2 and 3. | |
| LED current with RV1 at maximum (220 Ω) | 4.3 mA | 3.9 to 4.8 mA | Below 3.5 mA the LM317 may oscillate: check RV1 (220 Ω, not 250 Ω or more). | |

If a reading is out of range, stop and fix it before closing the box.

## 7. Calibration: jigs G1 to G3

Print the jigs before the final parts, with the same settings, lying flat and without supports. Try the real part in each hole, from the smallest to the largest, and pick the smallest one where it goes in and slides without too much play.

**G1 — Tube jig (calibracao-tubo.stl)**

| Series | Holes (mm) | Parameter to update |
|---|---|---|
| P2<br>Outer tube | 30.2 · 33.0 · 33.4 · 33.8 · 34.2 · 34.6 · 36.4 | tubeDiameter = chosen value − 0.2 |

**G2 — Parts jig (calibracao-pecas.stl)**

| Series | Holes (mm) | Parameter to update |
|---|---|---|
| L1<br>Slide (1.2 mm recess) | 26.2 · 26.4 · 26.6 · 26.8 · 27.0 | slideWidth = chosen value − 0.6 |
| SW1<br>Switch KCD1-101 | 20.8×14,6 · 21.0×14,8 · 21.2×15,0 · 21.4×15,2 · 21.6×15,4 | switchCutout (width and height) |
| D1<br>LED flange | 5.8 · 6.0 · 6.2 · 6.4 · 6.6 | ledFlange = chosen value − 0.4 |
| H1<br>M3 pilot hole | 2.4 · 2.6 · 2.8 | pilotHole (tight fit that still threads) |
| J1<br>Jack DC-022B | 7.6 · 7.8 · 8.0 · 8.2 | jackDiameter |
| RV1<br>Rheostat M7 thread | 7.0 · 7.3 · 7.5 · 7.8 | knobDiameter |

### Calibration record sheet

| Jig / series | Real part measured | Real size (mm) | Chosen hole (mm) | Date / person |
|---|---|---|---|---|
| P2<br>Outer tube | | | | |

| Jig / series | Real part measured | Real size (mm) | Chosen hole (mm) | Date / person |
|---|---|---|---|---|
| `L1`<br>Slide (1.2 mm recess) | | | | |
| `SW1`<br>Switch KCD1-101 | | | | |
| `D1`<br>LED flange | | | | |
| `H1`<br>M3 pilot hole | | | | |
| `J1`<br>Jack DC-022B | | | | |
| `RV1`<br>Rheostat M7 thread | | | | |

After choosing the values, update the parameters in holder.ts, regenerate the files and issue revision B.

## 8. Final inspection

| # | Check | OK | Not OK |
|---|---|---|---|
| 1 | The slide L1 moves in the channel without effort and without visible side play. | ☐ | ☐ |
| 2 | The pregoscope tube fits the ring and is held by M2 without cracking the PLA. | ☐ | ☐ |
| 3 | With the inner tube pulled out 19 mm on its scale the lens is about 0.8 mm from the slide and the image focuses. | ☐ | ☐ |
| 4 | The LED lights and the light is even over the window, with no hot spot. | ☐ | ☐ |
| 5 | LED current measured inside the range of the electrical test. | ☐ | ☐ |
| 6 | Lid closed with 4 screws, no trapped wire and no gap larger than 0.5 mm. | ☐ | ☐ |
| 7 | The switch snaps in; on v2 the knob turns smoothly and the plug seats. | ☐ | ☐ |
| 8 | Nothing loose when shaken gently; no burrs on the printed parts. | ☐ | ☐ |

| **Assembler** | | **Date** | | **Inspector** | |
|---|---|---|---|---|---|